

Flesch–Kincaid readability tests

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The **Flesch–Kincaid readability tests** are readability tests designed to indicate how difficult a reading passage in English is to understand. There are two tests, the Flesch reading ease, and the Flesch–Kincaid grade level. Although they use the same core measures (word length and sentence length), they have different weighting factors. The results of the two tests correlate approximately inversely: a text with a comparatively high score on the Reading Ease test should have a lower score on the grade-level test. Rudolf Flesch devised both systems, while J. Peter Kincaid developed the latter for the United States Navy.

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History

"The Flesch–Kincaid" (F–K) reading grade level was developed under contract to the US Navy in 1975 by J. Peter Kincaid and his team.^[1] Related UN Navy research directed by Kincaid delved into high-tech education (for example, the electronic authoring and delivery of technical information);^[2] usefulness of the Flesch–Kincaid readability formula;^[3] computer aids for editing tests;^[4] illustrated formats to teach procedures;^[5] and the Computer Readability Editing System (CRES).^[6]

The F–K formula was first used by the Army for assessing the difficulty of technical manuals in 1978 and soon after became the Department of Defense military standard. Pennsylvania was the first US state to require that automobile insurance policies be written at no higher than a ninth-grade level (14–15 years of age) of reading difficulty, as measured by the F–K formula. This is now a common requirement in many other states and for other legal documents such as insurance policies.^[3]

Flesch reading ease

In the Flesch reading-ease test, higher scores indicate material that is easier to read; lower numbers mark passages that are more difficult to read. The formula for the Flesch reading-ease score (FRES) test is

$$206.835 - 1.015 \left(\frac{\text{total words}}{\text{total sentences}} \right) - 84.6 \left(\frac{\text{total syllables}}{\text{total words}} \right) .^{[7]}$$

Scores can be interpreted as shown in the table below.

Score	Notes
90.0–100.0	easily understood by an average 11-year-old student
60.0–70.0	easily understood by 13- to 15-year-old students
0.0–30.0	best understood by university graduates

Reader's Digest magazine has a readability index of about 65, *Time* magazine scores about 52, an average 6th grade student's written assignment (age of 12) has a readability index of 60–70 (and a reading grade level of 6–7), and the *Harvard Law Review* has a general readability score in the low 30s. The highest (easiest) readability score possible is around 120 (e.g. every sentence consisting of only two one-syllable words; "The cat sat on the mat." scores 116). The score does not have a theoretical lower bound. It is possible to make the score as low as wanted by arbitrarily including words with many syllables. The sentence "This sentence, taken as a reading passage unto itself, is being used to prove a point." has a readability of 74.1. The sentence "The Australian platypus is seemingly a hybrid of a mammal and reptilian creature." scores 24.4 as it has 26 syllables and 13 words. While Amazon calculates the text of *Moby Dick* as 57.9,^[8] one particularly long sentence about sharks in chapter 64 has a readability score of −146.77.^[9] One sentence in the beginning of "Swann's Way", by Marcel Proust, has a score of −515.1.^[10]

The US Department of Defense uses the reading ease test as the standard test of readability for its documents and forms.^[11] Florida requires that life insurance policies have a Flesch reading ease score of 45 or greater.^[12]

Use of this scale is so ubiquitous that it is bundled with popular word processing programs and services such as KWord, IBM Lotus Symphony, Microsoft Office Word, WordPerfect, and WordPro.

Polysyllabic words affect this score significantly more than they do the grade level score.

Flesch–Kincaid grade level

These readability tests are used extensively in the field of education. The "Flesch–Kincaid Grade Level Formula" instead presents a score as a US grade level, making it easier for teachers, parents, librarians, and others to judge the readability level of various books and texts. It can also mean the number of years of education generally required to understand this text, relevant when the formula results in a number greater than 10. The grade level is calculated with the following formula:

$$0.39 \left(\frac{\text{total words}}{\text{total sentences}} \right) + 11.8 \left(\frac{\text{total syllables}}{\text{total words}} \right) - 15.59$$

The result is a number that corresponds with a U.S. grade level. The sentence, "The Australian platypus is seemingly a hybrid of a mammal and reptilian creature" is a 13.1 as it has 26 syllables and 13 words. The different weighting factors for words per sentence and syllables per word in each scoring system mean that the two schemes are not directly comparable and cannot be converted. The grade level formula emphasises sentence length over word length.

The lowest grade level score in theory is −3.40, but there are few real passages in which every sentence consists of a single one-syllable word. *Green Eggs and Ham* by Dr. Seuss comes close, averaging 5.7 words per sentence and 1.02 syllables per word, with a grade level of −1.3. (Most of the 50 used words are monosyllabic; "anywhere", which occurs eight times, is the only exception.)

See also

- Accessible publishing
- Automated readability index
- Coleman–Liau index
- Dale–Chall readability formula
- Gunning fog index
- Readability
- SMOG (simple measure of gobbledygook)

References

1. Kincaid, J.P., Fishburne, R.P., Rogers, R.L., & Chissom, B.S. (1975). Derivation of new readability formulas (automated readability index, fog count, and flesch reading ease formula) for Navy enlisted personnel. Research Branch Report 8–75. Chief of Naval Technical Training: Naval Air Station Memphis.
2. Kincaid JP, Braby R, Mears J (1988). "Electronic authoring and delivery of technical information". *Journal of Instructional Development* **11**: 8–13. doi:10.1007/bf02904998.
3. McClure G (1987). "Readability formulas: Useful or useless. (an interview with J. Peter Kincaid.)". *IEEE Transactions on Professional Communications* **30**: 12–15.
4. Kincaid JP, Braby R, Wulfeck WH II (1983). "Computer aids for editing tests". *Educational Technology* **23**: 29–33.
5. Braby R, Kincaid JP, Scott P, McDaniel W (1982). "Illustrated formats to teach procedures". *IEEE Transactions on Professional Communications* **25**: 61–66.
6. Kincaid JP, Aagard JA, O'Hara JW, Cottrell LK (1981). "Computer Readability Editing System". *IEEE Transactions on Professional Communications* **24** (1): 38–42. (also reported in *Aviation Week and Space Technology*, January 11, 1982, pp. 106–107.)
7. http://www.mang.canterbury.ac.nz/writing_guide/writing/flesch.shtml
8. Gabe Habash (July 20, 2011). "Book Lies: Readability is Impossible to Measure".
9. Melville, Herman. "Chapter 64: Stubb's Supper." *Moby Dick*. Lit2Go Edition. 1851. Web. <<http://etc.usf.edu/lit2go/42/moby-dick/745/chapter-64-stubbs-supper/>>. August 16, 2013.

"Though amid all the smoking horror and diabolism of a sea-fight, sharks will be seen longingly gazing up to the ship's decks, like hungry dogs round a table where red meat is being carved, ready to bolt down every killed man that is tossed to them; and though, while the valiant butchers over the deck-table are thus cannibally carving each other's live meat with carving-knives all gilded and tasselled, the sharks, also, with their jewel-hilted mouths, are quarrelsomely carving away under the table at the dead meat; and though, were you to turn the whole affair upside down, it would still be pretty much the same thing, that is to say, a shocking sharkish business enough for all parties; and though sharks also are the invariable outriders of all slave ships crossing the Atlantic, systematically trotting alongside, to be handy in case a parcel is to be carried anywhere, or a dead slave to be decently buried; and though one or two other like instances might be set down, touching the set terms, places, and occasions, when sharks do most socially congregate, and most hilariously feast; yet is there no conceivable time or occasion when you will find them in such countless numbers, and in gayer or more jovial spirits, than around a dead sperm whale, moored by night to a whaleship at sea."

10. Proust, Marcel. "Swann's Way." In *Search of Lost Time*. 2004. web (<http://www.gutenberg.org/ebooks/7178>). March 21, 2014.

"But I had seen first one and then another of the rooms in which I had slept during my life, and in the end I would revisit them all in the long course of my waking dream: rooms in winter, where on going to bed I would at once bury my head in a nest, built up out of the most diverse materials, the corner of my pillow, the top of my blankets, a piece of a shawl, the edge of my bed, and a copy of an evening paper, all of which things I would contrive, with the infinite patience of birds building their nests, to cement into one whole; rooms where, in a keen frost, I would feel the satisfaction of being shut in from the outer world (like the sea-swallow which builds at the end of a dark tunnel and is kept warm by the surrounding earth), and where, the fire keeping in all night, I would sleep wrapped up, as it were, in a great cloak of snug and savoury air, shot with the glow of the logs which would break out again in flame: in a sort of alcove without walls, a cave of warmth dug out of the heart of the room itself, a zone of heat whose boundaries were constantly shifting and altering in temperature as gusts of air ran across them to strike freshly upon my face, from the corners of the room, or from parts near the window or far from the fireplace which had therefore remained cold—or rooms in summer, where I would delight to feel myself a part of the warm evening, where the moonlight striking upon the half-opened shutters would throw down to the foot of my bed its enchanted ladder; where I would fall asleep, as it might be in the open air, like a titmouse which the breeze keeps poised in the focus of a sunbeam—or sometimes the Louis XVI room, so cheerful that I could never feel really unhappy, even on my first night in it: that room where the slender columns which lightly supported its ceiling would part, ever so gracefully, to indicate where the bed was and to keep it separate; sometimes again that little room with the high ceiling, hollowed in the form of a pyramid out of two separate storeys, and partly walled with mahogany, in which from the first moment my mind was drugged by the unfamiliar scent of flowering grasses, convinced of the hostility of the violet curtains and of the insolent indifference of a clock that chattered on at the top of its voice as though I were not there; while a strange and pitiless mirror with square feet, which stood across one corner of the room, cleared for itself a site I had not looked to find tenanted in the quiet surroundings of my normal field of vision: that room in which my mind, forcing itself for hours on end to leave its moorings, to elongate itself upwards so as to take on the exact shape of the room, and to reach to the summit of that monstrous funnel, had passed so many anxious nights while my body lay stretched out in bed, my eyes staring upwards, my ears straining, my nostrils sniffing uneasily, and my heart beating; until custom had changed the colour of the curtains, made the clock keep quiet, brought an expression of pity to the cruel, slanting face of the glass, disguised or even completely dispelled the scent of flowering grasses, and distinctly reduced the apparent loftiness of the ceiling."

11. Luo Si; et al. (November 5–10, 2001). *A statistical model for scientific readability*. Atlanta, GA, USA: CIKM '01.
12. "Readable Language in Insurance Policies" (<http://law.onecle.com/florida/insurance/627.4145.html>)

Further references

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External links

- PlainLanguage.com (<http://plainlanguage.com/newreadability.html>) – All About Readability
- The Principles of Readability by William H. DuBay (<http://www.impact-information.com/impactinfo/readability02.pdf>) – a brief introduction to readability research
- Flesh (<http://flesh.sourceforge.net/>) – free cross-platform, open source Java application that calculates the Flesch–Kincaid grade level and reading ease score

- **Style and Diction** (<http://www.gnu.org/software/diction/>) – free software by the GNU Project that calculates the Flesch–Kincaid readability score
- **The state of our union is ... dumber** (<http://www.guardian.co.uk/world/interactive/2013/feb/12/state-of-the-union-reading-level>) – Analysis of the State of the Union
- **readability.mackayst.com** (<http://readability.mackayst.com/>) – lists Flesch-Kincaid scores of Project Gutenberg books

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